OAVS · TGT COMPUTER TEACHER EXAMINATION

VB 6.0 — The Language

MCQ Question Bank

230 exam-oriented, non-repetitive questions with answers and short explanations

Syllabus covered (b. VB The language)

i. Variables, Constants, Data types, User-defined data types
ii. Scope of variables (Global, Local, Modular, Static)
iii. Control flow statements (If, IIf, Select Case)
iv. Looping and nesting (Do While, Do Until, For, For Each, Exit)
v. Arrays
vi. User-defined procedures and functions
vii. VB functions: Date, Mathematical, String, Conversion

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How to use: attempt each question first, then check the answer strip and explanation. Code is shown in monospace; assume default settings (`Option Base 0`, binary string comparison) unless stated. Date literals such as `#1/25/2024#` are month/day/year. Question types: Concept · Syntax · Output · Error · Scenario · Tricky · Match · Assertion–Reason · Statement.

SECTION 1

Variables, Constants and Data Types

Q1 - Q22
22 MCQs

Q1 | Concept | VB 6.0 — Variables

Which of the following determines the type of value a variable can store in Visual Basic 6.0?

A. Scope
B. Data type
C. Procedure
D. Loop

Ans: B

Explanation: A data type determines the kind of value a variable can hold and the operations applicable to it.

Q2 | Concept | VB 6.0 — Variables

A variable declared as `Dim x` (no `As` clause) has which data type?

A. String
B. Integer
C. Object
D. Variant

Ans: D

Explanation: Variant is the default type in VB 6.0. It can hold numbers, text or dates, and its subtype changes with the assigned value.

Q3 | Concept | VB 6.0 — Variables

Which statement forces every variable to be declared before it is used?

A. Option Explicit
B. Option Private Module
C. Option Compare Text
D. Option Base 1

Ans: A

Explanation: `Option Explicit` in the Declarations section turns an undeclared variable into a compile error (Variable not defined).

Q4 | Syntax | VB 6.0 — Variables

Which statement correctly declares an Integer variable?

A. Dim n As Integer
B. Declare n As Integer
C. Var n As Integer
D. Dim n Integer

Ans: A

Explanation: The syntax is `Dim variablename As datatype`.

Q5 | Concept | VB 6.0 — Variables

Which data type stores whole numbers from 0 to 255 in a single byte?

A. Integer
B. Boolean
C. Currency
D. Byte

Ans: D

Explanation: Byte is 1 byte (0–255). Integer is 2 bytes and Long is 4 bytes.

Q6 | Concept | VB 6.0 — Variables

Which data type is best suited to money calculations that need exactly four decimal places without rounding errors?

A. Single
B. Double
C. Currency
D. Date

Ans: C

Explanation: Currency is an 8-byte fixed-point type with 4 digits after the decimal point, so it avoids floating-point rounding errors.

Q7 | Concept | VB 6.0 — Variables

The type-declaration character `@` after a variable name (for example `Price@`) denotes:

A. Currency
B. Byte
C. Boolean
D. Single

Ans: A

Explanation: Suffixes: % Integer, & Long, ! Single, # Double, @ Currency, $ String.

Q8 | Syntax | VB 6.0 — Variables

Which statement correctly declares a symbolic constant?

A. Const Pi = 3.14159
B. Dim Const Pi = 3.14159
C. Constant Pi = 3.14159
D. Const Pi As = 3.14159

Ans: A

Explanation: `Const name = value` declares a constant whose value cannot be changed while the program runs.

Q9 | Concept | VB 6.0 — Variables

Built-in names such as `vbCrLf`, `vbYes` and `vbRed` are examples of:

A. Global variables
B. Intrinsic constants
C. User-defined constants
D. Static variables

Ans: B

Explanation: VB provides intrinsic (built-in) constants so that code need not use raw numbers.

Q10 | Concept | VB 6.0 — Variables

An uninitialised numeric variable, for example `Dim n As Integer`, holds the value:

A. 0
B. -1
C. 1
D. Null

Ans: A

Explanation: Numeric variables start at 0, Strings at an empty string, Booleans at False and Variants at Empty.

Q11 | Concept | VB 6.0 — Variables

What is the numeric value of the Boolean constant `True` in VB 6.0?

A. 0
B. 1
C. 255
D. -1

Ans: D

Explanation: True is stored as -1 (all bits set) and False as 0. Any non-zero value converts to True.

Q12 | Syntax | VB 6.0 — Variables

Which declaration creates a fixed-length string of 10 characters?

A. Dim s(10) As String
B. Dim s As String[10]
C. Dim s As String * 10
D. Dim s As String(10)

Ans: C

Explanation: `String * n` fixes the length. `Dim s(10) As String` would instead create an array of 11 ordinary strings.

Q13 | Concept | VB 6.0 — Variables

Which of the following is a valid VB 6.0 variable name?

A. Total_1
B. Total-1
C. 1Total
D. Total 1

Ans: A

Explanation: A name must begin with a letter and contain only letters, digits and underscores. It cannot contain spaces or symbols such as `-`.

Q14 | Tricky | VB 6.0 — Variables

In the statement `Dim a, b As Integer`, the variable `a` is of type:

A. Boolean
B. Integer
C. Variant
D. Long

Ans: C

Explanation: VB 6.0 needs an `As` clause for each variable. Only `b` is Integer; `a` defaults to Variant.

Q15 | Concept | VB 6.0 — Variables

The data type that stores both date and time values, internally as a floating-point number, is:

A. Variant only
B. String
C. Date
D. Long

Ans: C

Explanation: The Date type (8 bytes) stores the date as the integer part and the time of day as the fractional part.

Q16 | Concept | VB 6.0 — Variables

Which `VarType` return value indicates a Boolean?

A. 2
B. 11
C. 7
D. 8

Ans: B

Explanation: VarType codes: 2 Integer, 3 Long, 4 Single, 5 Double, 7 Date, 8 String, 11 Boolean (constant `vbBoolean`).

Q17 | Tricky | VB 6.0 — Variables

If `v` is a Variant containing Null, what does `IsNull(v + 5)` return?

```
Dim v As Variant
v = Null
MsgBox IsNull(v + 5)
```

A. Run-time error
B. It returns 5
C. True
D. False

Ans: C

Explanation: Null propagates through arithmetic: any expression with Null as an operand gives Null.

Q18 | Syntax | VB 6.0 — Variables

Which statement releases an object variable's reference to its object?

A. Set obj = Nothing
B. Erase obj
C. obj = Empty
D. Const obj = 0

Ans: A

Explanation: `Nothing` is assigned to object variables with `Set`. It means the variable no longer refers to any object.

Q19 | Concept | VB 6.0 — Variables

`Option Compare Text`, placed in the Declarations section, makes string comparisons:

A. Insensitive to spaces
B. Numeric only
C. Case-sensitive
D. Case-insensitive

Ans: D

Explanation: The default (`Option Compare Binary`) is case-sensitive. Text mode ignores case.

Q20 | Concept | VB 6.0 — Variables

In the common naming convention, which prefix suits a Long variable?

A. sng
B. int
C. str
D. lng

Ans: D

Explanation: Typical prefixes: int (Integer), lng (Long), sng (Single), dbl (Double), str (String), bln (Boolean).

Q21 | Tricky | VB 6.0 — Variables

Which statement about the Decimal type in VB 6.0 is correct?

A. It is a 2-byte integer type

B. It exists only as a Variant subtype, created with CDec

C. It is the default data type

D. `Dim d As Decimal` is the standard declaration

Ans: B

Explanation: VB 6.0 cannot declare a variable As Decimal. Decimal values are obtained by converting a Variant with `CDec`.

Q22 | Error | VB 6.0 — Variables

What happens when this code runs?

```
Dim a As Integer
a = 20000
MsgBox a + a
```

A. -25536 is displayed

B. 40000 is displayed

C. Run-time error 6: Overflow

D. 32767 is displayed

Ans: C

Explanation: Integer + Integer is calculated as Integer in VB 6.0 and 40000 exceeds 32,767. Use `CLng(a) + a` to avoid it.

SECTION 2

User-Defined Data Types

Q23 - Q30
8 MCQs

Q23 | Concept | VB 6.0 — User-Defined Types

Which pair of statements defines a user-defined data type in VB 6.0?

A. Class ... End Class

B. Type ... End Type

C. Enum ... End Enum

D. Record ... End Record

Ans: B

Explanation: `Type … End Type` groups related elements of different data types under one name. `Enum` defines named integer constants only.

Q24 | Syntax | VB 6.0 — User-Defined Types

Given the definition below, which statement correctly assigns 85 to the Marks member?

```
Type Student
    Name As String
    Marks As Integer
End Type
Dim s As Student
```

A. s.Marks = 85

B. Student.Marks = 85

C. s!Marks = 85

D. s(Marks) = 85

Ans: A

Explanation: Members of a UDT variable are accessed with the dot operator: `variable.member`.

Q25 | Concept | VB 6.0 — User-Defined Types

Where must a `Type` statement be placed?

A. Inside a Function only
B. After the last Next statement
C. In the Declarations section of a module, outside any procedure
D. Inside any Sub procedure

Ans: C

Explanation: A Type block is a module-level declaration and cannot appear inside a procedure.

Q26 | Concept | VB 6.0 — User-Defined Types

A user-defined type declared inside a Form module must be declared as:

A. Private
B. Global
C. Public
D. Static

Ans: A

Explanation: Public UDTs are allowed only in standard (.bas) modules. In a form module the Type must be Private.

Q27 | Output | VB 6.0 — User-Defined Types

What is displayed by the following code?

```
Type Pt
    X As Integer
    Y As Integer
End Type

Dim p As Pt
p.X = 4: p.Y = 6
MsgBox p.X * p.Y
```

A. 10
B. 24
C. 0
D. 46

Ans: B

Explanation: p.X × p.Y = 4 × 6 = 24.

Q28 | Concept | VB 6.0 — User-Defined Types

What is the main purpose of a user-defined data type?

A. To declare a constant value
B. To group related data of different types under one name
C. To store many values of one type using an index
D. To define reusable code that returns a value

Ans: B

Explanation: A UDT works like a record, for example a student with Name (String), Roll (Integer) and Fee (Currency).

Q29 | Syntax | VB 6.0 — User-Defined Types

Which statement lets you refer to many members of a UDT variable without repeating the variable name?

A. Select Case … End Select
B. For Each … Next
C. Do … Loop
D. With … End With

Ans: D

Explanation: `With s … End With` allows `.Name` and `.Marks` to be written without the prefix `s`.

Q30 | Concept | VB 6.0 — User-Defined Types

If `s1` and `s2` are variables of the same user-defined type, what does `s2 = s1` do?

A. Causes a type-mismatch error
B. Copies the values of all members of s1 into s2
C. Makes s2 an alias of s1
D. Copies only the first member

Ans: B

Explanation: Assignment between UDT variables of the same type copies every member (a value copy, not a reference).

SECTION 3

Scope of Variables

Q31 - Q44
14 MCQs

Q31 | Concept | VB 6.0 — Scope

A variable declared with `Dim` inside a procedure has which scope?

A. Local (procedure-level)
B. Module-level
C. Global (project-wide)
D. Static only

Ans: A

Explanation: A local variable is visible only inside its own procedure and is destroyed when the procedure ends.

Q32 | Concept | VB 6.0 — Scope

Which keyword, used in the Declarations section of a standard module, makes a variable available to the entire project?

A. Public
B. Private
C. Static
D. Dim

Ans: A

Explanation: `Public` (older form: `Global`) in a standard module gives project-wide scope.

Q33 | Concept | VB 6.0 — Scope

A variable declared with `Dim` in the General Declarations section of a form is:

A. Destroyed after each procedure call
B. Available to all procedures of that form only
C. Local to the first procedure
D. Available to the entire project

Ans: B

Explanation: This is module-level scope: visible to every procedure in that module and nowhere else.

Q34 | Syntax | VB 6.0 — Scope

The `Global` keyword for declaring a variable is allowed only in the:

A. Declarations section of a standard module
B. Private section of a Type
C. Event code of a form
D. Body of any procedure

Ans: A

Explanation: `Global` is the older equivalent of `Public` and works only in the Declarations section of a standard module.

Q35 | Concept | VB 6.0 — Scope

Which keyword makes a local variable keep its value between calls to the procedure?

A. Private
B. Dim
C. Static
D. Const

Ans: C

Explanation: A `Static` variable keeps its value after the procedure ends, but its scope remains local.

Q36 | Output | VB 6.0 — Scope

`Counter` is called three times. What does the MsgBox show on the third call?

```
Sub Counter()
    Static n As Integer
    n = n + 1
    MsgBox n
End Sub
```

A. 1
B. 2
C. 3
D. 0

Ans: C

Explanation: `Static n` retains its value across calls (1, 2, 3). With `Dim n` it would show 1 every time.

Q37 | Output | VB 6.0 — Scope

What is displayed by `MsgBox x` in `Main`?

```
Dim x As Integer          ' module level

Sub Test()
    Dim x As Integer
    x = 50
End Sub

Sub Main()
    x = 10
    Test
    MsgBox x
End Sub
```

A. 60
B. 10
C. 0
D. 50

Ans: B

Explanation: The local `x` inside `Test` hides the module-level `x`, so the module-level x stays 10.

Q38 | Error | VB 6.0 — Scope

Procedure A declares `Dim total` and procedure B uses `total` without declaring it. With Option Explicit on, the result is:

A. B reads the value stored by A
B. Run-time error 13: Type mismatch
C. Compile error: Variable not defined
D. B gets 0 silently

Ans: C

Explanation: Local variables are invisible outside their own procedure, so B's `total` is undeclared.

Q39 | Syntax | VB 6.0 — Scope

A Public variable `Count` is declared in Form1. How is it accessed from code in Form2?

A. Form1.Count
B. Count
C. Form1!Count
D. Public.Count

Ans: A

Explanation: A Public variable declared in a form behaves like a property of that form and is referenced as `FormName.Variable`.

Q40 | Concept | VB 6.0 — Scope

What is the effect of writing `Static` before `Sub` (for example `Static Sub Calc()`)?

A. The procedure cannot be called twice
B. The procedure becomes a Function
C. The procedure is visible to the whole project
D. All local variables in the procedure become static

Ans: D

Explanation: Declaring a procedure Static preserves every local variable in it between calls.

Q41 | Concept | VB 6.0 — Scope

At module level in the Declarations section, `Dim x As Integer` is equivalent to:

A. Private x As Integer
B. Static x As Integer
C. Public x As Integer
D. Global x As Integer

Ans: A

Explanation: Both give module-level scope, visible only within that module.

Q42 | Concept | VB 6.0 — Scope

A Public variable declared in a standard module keeps its value until:

A. The next statement executes
B. The application ends
C. The procedure that uses it ends
D. The first form is closed

Ans: B

Explanation: Public (global) variables live for the entire run of the program.

Q43 | Concept | VB 6.0 — Scope

Arrange the scopes from narrowest to widest.

A. Local → Public → Module-level
B. Public → Module-level → Local
C. Module-level → Local → Public
D. Local → Module-level → Public (project-wide)

Ans: D

Explanation: A procedure-level variable is visible in one procedure, module-level in one module and Public across the project.

Q44 | Error | VB 6.0 — Scope

What happens when this code is compiled?

```
Sub Test()
    Dim x As Integer
    Dim x As String
End Sub
```

A. Run-time error 13
B. x becomes a Variant
C. Compile error: Duplicate declaration in current scope
D. The second Dim silently replaces the first

Ans: C

Explanation: The same name cannot be declared twice at the same scope level.

SECTION 4

Control Flow Statements

Q45 – Q62
18 MCQs

Q45 | Syntax | VB 6.0 — Control Flow

Which keyword adds a further condition in a block `If` statement?

A. ElseIf
B. Else When
C. Elsif
D. Elif

Ans: A

Explanation: Syntax: `If … Then … ElseIf … Then … Else … End If`.

Q46 | Output | VB 6.0 — Control Flow

What is displayed by the following code?

```
x = 15
If x > 10 Then
    MsgBox "A"
ElseIf x > 5 Then
    MsgBox "B"
Else
    MsgBox "C"
End If
```

A. C
B. A
C. A then B
D. B

Ans: B

Explanation: The first true condition (x > 10) runs and the rest of the block is skipped.

Q47 | Syntax | VB 6.0 — Control Flow

Which expression returns "Pass" when `m` is 40 or more, otherwise "Fail"?

A. IIf("Pass", "Fail", m >= 40)
B. IIf(m >= 40) Then "Pass"
C. IIf(m >= 40, "Fail", "Pass")
D. IIf(m >= 40, "Pass", "Fail")

Ans: D

Explanation: Syntax: `IIf(expression, truepart, falsepart)`. It is an inline function, not a block statement.

Q48 | Error | VB 6.0 — Control Flow

If `n = 0`, what happens when the following expression is evaluated?

```
r = IIf(n <> 0, 100 / n, 0)
```

A. Run-time error 11: Division by zero
B. It returns Null
C. It returns 100
D. It returns 0

Ans: A

Explanation: IIf evaluates both the truepart and the falsepart before choosing one, so `100 / n` is evaluated anyway.

Q49 | Syntax | VB 6.0 — Control Flow

Which `Case` clause tests whether a value lies from 1 to 5?

A. Case 1 - 5
B. Case 1 To 5
C. Case 1..5
D. Case Between 1 And 5

Ans: B

Explanation: `Case a To b` tests an inclusive range.

Q50 | Syntax | VB 6.0 — Control Flow

Which `Case` clause matches scores of 90 and above?

A. Case >= 90
B. Case If >= 90
C. Case Is >= 90
D. Case 90+

Ans: C

Explanation: A relational test in `Case` needs the `Is` keyword: `Case Is >= 90`.

Q51 | Output | VB 6.0 — Control Flow

What is the value of `s` after the following code runs?

```
n = 3
Select Case n
    Case 1 To 5
        s = "Low"
    Case 3
        s = "Three"
    Case Else
        s = "Other"
End Select
```

A. "Three"
B. "LowThree"
C. "Low"
D. "Other"

Ans: C

Explanation: Select Case executes only the first matching Case. 3 lies in 1 To 5, so `Case 3` is never reached.

Q52 | Concept | VB 6.0 — Control Flow

In `Select Case`, the `Case Else` clause runs when:

A. No other Case clause matches
B. The expression is Null only
C. Every Case clause matches
D. The first Case fails to compile

Ans: A

Explanation: `Case Else` is optional. It acts as the default branch.

Q53 | Tricky | VB 6.0 — Control Flow

Which `Case` clause matches the values 2, 4 and 6?

A. Case 2 : 4 : 6
B. Case 2 And 4 And 6
C. Case 2 Or 4 Or 6
D. Case 2, 4, 6

Ans: D

Explanation: List values separated by commas. `2 Or 4 Or 6` is evaluated bitwise to 6, so only 6 would match.

Q54 | Output | VB 6.0 — Control Flow

What grade is assigned by the following code?

```
m = 72
Select Case True
    Case m >= 90: g = "A"
    Case m >= 60: g = "B"
    Case Else:    g = "C"
End Select
```

A. "C"
B. "B"
C. "A"
D. No assignment

Ans: B

Explanation: `Select Case True` runs the first Case whose condition is True. 72 >= 90 is False, but 72 >= 60 is True.

Q55 | Tricky | VB 6.0 — Control Flow

With `a = 0`, what happens when this statement executes?

```
If a <> 0 And 10 / a > 1 Then MsgBox "OK"
```

A. Compile error
B. The Then part runs
C. Run-time error 11 (Division by zero)
D. The condition is False and the If is skipped

Ans: C

Explanation: VB 6.0 does not short-circuit `And`. Both operands are always evaluated, so `10 / a` raises an error.

Q56 | Syntax | VB 6.0 — Control Flow

Which is a valid single-line If statement?

A. If (x > 5) { MsgBox "Big" }
B. If x > 5 : MsgBox "Big"
C. If x > 5 Then MsgBox "Big"
D. If x > 5 Then MsgBox "Big" End If

Ans: C

Explanation: A single-line If has no `End If`. Adding `End If` to it is a syntax error.

Q57 | Output | VB 6.0 — Control Flow

What is the value of `r`?

```
a = 5: b = 8
If Not (a > b) And b > 7 Then r = "X" Else r = "Y"
```

A. "X"
B. No value is assigned
C. "XY"
D. "Y"

Ans: A

Explanation: `Not (a > b)` is True and `b > 7` is True, so `And` gives True and the Then part runs.

Q58 | Output | VB 6.0 — Control Flow

What does the MsgBox display?

```
MsgBox (5 > 3) Xor (2 > 1)
```

A. False
B. Run-time error
C. True
D. Null

Ans: A

Explanation: `Xor` is True only when exactly one operand is True. Both operands here are True.

Q59 | Tricky | VB 6.0 — Control Flow

What is the value of `Not 5` in VB 6.0?

A. -5
B. -6
C. 0
D. False

Ans: B

Explanation: On numbers `Not` works bitwise (it inverts every bit), so Not 5 = -(5 + 1) = -6. It gives True/False only for Boolean operands.

Q60 | Output | VB 6.0 — Control Flow

What is the value of `m` after the code runs?

```
a = 10: b = 20: c = 15
If a > b Then
    m = a
ElseIf b > c Then
    m = b
Else
    m = c
End If
```

A. 0
B. 15
C. 20
D. 10

Ans: C

Explanation: a > b is False, so the ElseIf is tested. b > c is True and m = b = 20.

Q61 | Tricky | VB 6.0 — Control Flow

With default (binary) comparison, what is `t` after this code?

```
d = "sat"
Select Case d
    Case "Sat", "Sun": t = "Weekend"
    Case Else: t = "Weekday"
End Select
```

A. "Weekday"
B. "Weekend"
C. Empty
D. Run-time error

Ans: A

Explanation: String comparison in `Case` is case-sensitive by default, so "sat" does not match "Sat".

Q62 | Tricky | VB 6.0 — Control Flow

What is the value of `Not 3 > 2 Or 1 > 2`?

A. True
B. Null
C. Run-time error
D. False

Ans: D

Explanation: Precedence: comparison, then Not, And, Or. (3 > 2) = True, Not gives False, 1 > 2 is False, and False Or False = False.

SECTION 5

Looping and Nested Loops

Q63 - Q85
23 MCQs

Q63 | Concept | VB 6.0 — Loops

Which loop tests its condition at the end and therefore always runs at least once?

A. Do Until ... Loop
B. Do While ... Loop
C. For ... Next with a False limit
D. Do ... Loop While

Ans: D

Explanation: When the condition follows `Loop`, the body executes once before the test.

Q64 | Output | VB 6.0 — Loops

What is the final value of `s`?

```
i = 1
Do While i <= 3
    s = s & i
    i = i + 1
Loop
```

A. "123"
B. "1234"
C. "12"
D. "6"

Ans: A

Explanation: The loop runs for i = 1, 2, 3, and `&` appends each digit to the string.

Q65 | Output | VB 6.0 — Loops

What does the MsgBox display?

```
n = 10
Do Until n < 5
    n = n - 3
Loop
MsgBox n
```

A. 7
B. 1
C. 4
D. 10

Ans: C

Explanation: n goes 10 → 7 → 4. The loop stops when `n < 5` becomes True, with n = 4.

Q66 | Output | VB 6.0 — Loops

What does the MsgBox display?

```
x = 10
Do
    x = x + 1
Loop While x < 5
MsgBox x
```

A. Nothing (infinite loop)
B. 11
C. 10
D. 5

Ans: B

Explanation: The body runs once (x = 11) before `x < 5` is tested. The test fails, so the loop ends.

Q67 | Output | VB 6.0 — Loops

What is the final value of `s`?

```
For i = 10 To 1 Step -3
    s = s & i & " "
Next i
```

A. The loop does not execute
B. "10 8 6 4 2"
C. "10 7 4"
D. "10 7 4 1"

Ans: D

Explanation: With Step -3 the counter takes the values 10, 7, 4, 1 and stops when it would go below 1.

Q68 | Concept | VB 6.0 — Loops

How many times does this loop body execute?

```
For i = 1 To 20 Step 5
    …
Next i
```

A. 4
B. 20
C. 5
D. 3

Ans: A

Explanation: The counter takes the values 1, 6, 11 and 16, and 21 exceeds 20.

Q69 | Tricky | VB 6.0 — Loops

After the loop finishes normally, the value of `i` is:

```
For i = 1 To 5
    …
Next i
```

A. 1
B. 0
C. 5
D. 6

Ans: D

Explanation: After the last pass the counter is incremented once more to 6 and the test fails. Control leaves the loop with i = 6.

Q70 | Concept | VB 6.0 — Loops

The loop `For k = 5 To 1` (no Step) executes its body:

A. Infinitely
B. Zero times
C. Once
D. Five times

Ans: B

Explanation: The default Step is +1. Since the start value 5 is already greater than the end value 1, the loop is skipped.

Q71 | Output | VB 6.0 — Loops

What does the MsgBox display?

```
For i = 1 To 10
    If i * i > 30 Then Exit For
Next i
MsgBox i
```

A. 11
B. 5
C. 6
D. 10

Ans: C

Explanation: At i = 6, i*i = 36 > 30, so Exit For runs and i stays 6.

Q72 | Concept | VB 6.0 — Loops

Which loop visits every element of an array or collection without using an index counter?

A. For … Step
B. For Each … Next
C. While … Wend
D. Do Until … Loop

Ans: B

Explanation: `For Each element In group` traverses each member of an array or collection in turn.

Q73 | Error | VB 6.0 — Loops

What is wrong with this code?

```
Dim a(2) As Integer, n As Integer
For Each n In a
    …
Next
```

A. Next must carry the variable name
B. The array must be dynamic
C. For Each cannot be used with arrays
D. The loop variable of For Each over an array must be a Variant

Ans: D

Explanation: Over an array, the For Each control variable must be Variant. Over a collection it may be Variant or Object.

Q74 | Output | VB 6.0 — Loops

What is the value of `c` after the nested loops finish?

```
For i = 1 To 3
    For j = 1 To 2
        c = c + 1
    Next j
Next i
```

A. 6
B. 5
C. 9
D. 3

Ans: A

Explanation: The outer loop runs 3 times and the inner loop 2 times per pass, so 3 × 2 = 6.

Q75 | Tricky | VB 6.0 — Loops

What is the value of `c` after this code runs?

```
For i = 1 To 3
    For j = 1 To 3
        If j = 2 Then Exit For
        c = c + 1
    Next j
Next i
```

A. 3
B. 1
C. 9
D. 6

Ans: A

Explanation: `Exit For` leaves only the inner loop. For each of the 3 outer passes, c is incremented once (j = 1).

Q76 | Output | VB 6.0 — Loops

How many asterisks are in `s` at the end?

```
For i = 1 To 3
    For j = 1 To i
        s = s & "*"
    Next j
Next i
```

A. 3
B. 6
C. 5
D. 9

Ans: B

Explanation: The inner loop runs i times: 1 + 2 + 3 = 6.

Q77 | Tricky | VB 6.0 — Loops

Which of these exit statements does NOT exist in VB 6.0?

A. Exit For
B. Exit Do
C. Exit Sub
D. Exit While

Ans: D

Explanation: `While … Wend` has no Exit statement. Use `Do … Loop` when early exit is needed.

Q78 | Output | VB 6.0 — Loops

What does the MsgBox display?

```
n = 0
Do
    n = n + 2
    If n > 6 Then Exit Do
Loop
MsgBox n
```

A. 6
B. 10
C. 7
D. 8

Ans: D

Explanation: n takes 2, 4, 6, 8. At 8, `n > 6` is True and Exit Do ends the loop.

Q79 | Error | VB 6.0 — Loops

Why does this loop never terminate?

```
i = 1
Do While i <= 5
    s = s & i
Loop
```

A. Do While cannot use <=
B. The loop needs a Wend
C. The counter i is never changed inside the loop
D. s must be declared as Long

Ans: C

Explanation: The condition `i <= 5` stays True forever because nothing increments i.

Q80 | Output | VB 6.0 — Loops

What does the MsgBox display?

```
n = 1
While n < 20
    n = n * 2
Wend
MsgBox n
```

A. 16
B. 64
C. 20
D. 32

Ans: D

Explanation: n goes 1, 2, 4, 8, 16, 32. The loop ends when n < 20 is False.

Q81 | Output | VB 6.0 — Loops

What does the MsgBox display?

```
i = 5
Do
    i = i - 1
Loop Until i <= 5
MsgBox i
```

A. 5 **B.** 4

C. 1 **D.** 0

Ans: B

Explanation: The body runs once (i = 4) and then `i <= 5` is True, so the loop ends. The Until test comes after the body.

Q82 | Output | VB 6.0 — Loops

What is the final value of `t`?

```
Dim c As New Collection, v As Variant, t As Integer
c.Add 5: c.Add 10: c.Add 15
For Each v In c
    t = t + v
Next
```

A. 3 **B.** 15

C. 10 **D.** 30

Ans: D

Explanation: For Each visits each item of the collection: 5 + 10 + 15 = 30.

Q83 | Concept | VB 6.0 — Loops

How many times does this loop execute?

```
For x = 1 To 2 Step 0.5
    …
Next x
```

A. 2 **B.** 5

C. 3 **D.** 4

Ans: C

Explanation: x takes 1, 1.5 and 2. Step values may be fractional.

Q84 | Tricky | VB 6.0 — Loops

What is the final value of `s`?

```
For i = 1 To 5
    i = i + 1
    s = s & i
Next i
```

A. "2468" **B.** "135"

C. "246" **D.** "12345"

Ans: C

Explanation: Changing the counter inside the loop affects the iteration: i = 2, then 4, then 6, and the loop ends.

Q85 | Concept | VB 6.0 — Loops

A `Do … Loop` with no `While` or `Until` condition runs:

A. Until an Exit Do (or other exit statement) is executed
B. Never
C. Until the counter reaches 10
D. Exactly once

Ans: A

Explanation: Without a condition the loop is infinite unless Exit Do, Exit Sub or Exit Function is used.

SECTION 6

Arrays

Q86 - Q103
18 MCQs

Q86 | Concept | VB 6.0 — Arrays

How many elements are created by `Dim a(5) As Integer` (default settings)?

A. 4
B. 7
C. 6
D. 5

Ans: C

Explanation: The default lower bound is 0, so the indexes run from 0 to 5, giving six elements.

Q87 | Concept | VB 6.0 — Arrays

If `Option Base 1` is specified, how many elements does `Dim a(5)` create?

A. 4
B. 6
C. 5
D. 0

Ans: C

Explanation: With Option Base 1 the indexes run from 1 to 5.

Q88 | Tricky | VB 6.0 — Arrays

How many elements does `Dim m(5 To 12) As Integer` have?

A. 9
B. 7
C. 12
D. 8

Ans: D

Explanation: Elements = upper − lower + 1 = 12 − 5 + 1 = 8.

Q89 | Concept | VB 6.0 — Arrays

`Dim t(2, 3) As Integer` creates a two-dimensional array with how many elements?

A. 6
B. 5
C. 12
D. 8

Ans: C

Explanation: Rows 0–2 (3) × columns 0–3 (4) = 12.

Q90 | Concept | VB 6.0 — Arrays

Which keyword retains the existing data while a dynamic array is resized?

A. Preserve
B. Keep
C. Static
D. Retain

Ans: A

Explanation: `ReDim Preserve a(n)` keeps current values. A plain `ReDim` clears them.

Q91 | Tricky | VB 6.0 — Arrays

When `ReDim Preserve` is used on a multi-dimensional array, which dimension can be changed?

A. Any dimension
B. None of them
C. Only the first dimension
D. Only the last dimension

Ans: D

Explanation: Preserve restricts resizing to the last dimension (and the array's data type cannot be changed).

Q92 | Output | VB 6.0 — Arrays

What does the MsgBox display?

```
Dim a(4) As Integer, i As Integer
For i = 0 To 4
    a(i) = i * 2
Next i
MsgBox a(1) + a(4)
```

A. 6
B. 10
C. 8
D. 12

Ans: B

Explanation: a(1) = 2 and a(4) = 8, so 2 + 8 = 10.

Q93 | Error | VB 6.0 — Arrays

What happens at run time when this code executes?

```
Dim a(3) As Integer
a(4) = 1
```

A. Compile error: Array already dimensioned
B. Run-time error 6: Overflow
C. The array grows automatically
D. Run-time error 9: Subscript out of range

Ans: D

Explanation: `Dim a(3)` allows indexes 0–3 only. Index 4 is outside the bounds.

Q94 | Concept | VB 6.0 — Arrays

What does `UBound(v)` return for `v = Array(10, 20, 30)` (default Option Base)?

```
Dim v As Variant
v = Array(10, 20, 30)
```

A. 1
B. 3
C. 2
D. 30

Ans: C

Explanation: `Array()` gives a zero-based Variant array, so the upper bound is index 2 for three elements.

Q95 | Concept | VB 6.0 — Arrays

What does the `Erase` statement do to a dynamic array?

A. Releases the storage allocated to it
B. Reverses the order of its elements
C. Sorts its elements
D. Resizes it to one element

Ans: A

Explanation: Erase frees a dynamic array. On a fixed-size array it resets all elements to their defaults.

Q96 | Output | VB 6.0 — Arrays

What is the value of `UBound(a, 2)`?

```
Dim a(3, 5) As Integer
```

A. 3
B. 4
C. 6
D. 5

Ans: D

Explanation: The second argument selects the dimension. The second dimension of `a(3, 5)` has upper bound 5.

Q97 | Tricky | VB 6.0 — Arrays

What does the MsgBox display?

```
Dim a() As Integer
ReDim a(2)
a(1) = 9
ReDim a(4)
MsgBox a(1)
```

A. 9
B. 4
C. 0
D. Run-time error

Ans: C

Explanation: A `ReDim` without `Preserve` reinitialises the array, so a(1) is 0 again.

Q98 | Error | VB 6.0 — Arrays

What is the result of the `ReDim` statement below?

```
Dim a(5) As Integer
ReDim a(10)
```

A. The array is cleared but stays at 6 elements
B. The array is resized to 11 elements
C. Run-time error 9
D. Compile error: Array already dimensioned

Ans: D

Explanation: ReDim works only on dynamic arrays (`Dim a()`). A fixed-size array cannot be redimensioned.

Q99 | Syntax | VB 6.0 — Arrays

Given a user-defined type `Emp` with member `Id`, which expression refers to the Id of the SECOND element of `Dim e(2) As Emp`?

A. e(Id)(1)
B. e(2).Id
C. e.Id(1)
D. e(1).Id

Ans: D

Explanation: Indexes start at 0, so the second element is e(1). e(2) is the third.

Q100 | Syntax | VB 6.0 — Arrays

Which procedure header accepts an Integer array as a parameter?

A. Sub Show(a As Integer)
B. Sub Show(a[] As Integer)
C. Sub Show(a() As Integer)
D. Sub Show(ByVal a() As Integer)

Ans: C

Explanation: Array parameters use empty parentheses. Arrays are always passed by reference, so ByVal is not allowed.

Q101 | Output | VB 6.0 — Arrays

What is `LBound(x, 1) + UBound(x, 2)`?

```
Dim x(1 To 3, 0 To 4) As Integer
```

A. 3
B. 7
C. 4
D. 5

Ans: D

Explanation: LBound of dimension 1 is 1 and UBound of dimension 2 is 4, so 1 + 4 = 5.

Q102 | Tricky | VB 6.0 — Arrays

What does `UBound(Filter(Array("apple", "banana", "cherry"), "an"))` return?

A. -1
B. 0
C. 2
D. 1

Ans: B

Explanation: Filter returns only the elements containing "an", which is just "banana". It is a zero-based array with one element, so UBound = 0.

Q103 | Output | VB 6.0 — Arrays

What does `Join(Array("A", "B", "C"), "-")` return?

A. "ABC"
B. "-A-B-C"
C. "A-B-C"
D. "A,B,C"

Ans: C

Explanation: Join concatenates the array elements using the given delimiter (the reverse of Split).

SECTION 7

User-Defined Procedures and Functions

Q104 – Q123
20 MCQs

Q104 | Concept | VB 6.0 — Procedures

What is the main difference between a Sub procedure and a Function procedure?

A. A Sub can take arguments; a Function cannot
B. A Sub must be called with Call; a Function cannot
C. A Function cannot be Public; a Sub can
D. A Function returns a value; a Sub does not

Ans: D

Explanation: Both can take arguments. A Function returns a value and can be used inside expressions.

Q105 | Syntax | VB 6.0 — Procedures

In VB 6.0, how does a Function return its result?

A. By declaring an Out parameter
B. By using a Return statement
C. By using a Yield statement
D. By assigning the value to the function name

Ans: D

Explanation: Example: `Add = a + b` inside `Function Add(…)`. VB 6.0 has no `Return value` statement.

Q106 | Syntax | VB 6.0 — Procedures

For `Sub Show(n As Integer)`, which call is correct?

A. Show := 5
B. Call Show(5)
C. Execute Show(5)
D. Call Show 5

Ans: B

Explanation: `Call Show(5)` needs parentheses, while `Show 5` (without Call) does not.

Q107 | Concept | VB 6.0 — Procedures

By default, arguments in VB 6.0 procedures are passed:

A. By constant
B. By copy-restore
C. By value (ByVal)
D. By reference (ByRef)

Ans: D

Explanation: ByRef is the default, so the procedure can change the caller’s variable.

Q108 | Output | VB 6.0 — Procedures

What is displayed by the MsgBox?

```
Sub Change(ByVal a As Integer, b As Integer)
    a = a + 10
    b = b + 10
End Sub

Dim x As Integer, y As Integer
x = 1: y = 1
Change x, y
MsgBox x & "," & y
```

A. "1,1"
B. "11,11"
C. "1,11"
D. "11,1"

Ans: C

Explanation: a is ByVal (a copy), so x is unchanged. b is ByRef by default, so y becomes 11.

Q109 | Output | VB 6.0 — Procedures

What does the MsgBox display?

```
Function Sq(n As Integer) As Integer
    Sq = n * n
End Function

MsgBox Sq(3) + Sq(2)
```

A. 13
B. 25
C. 10
D. 5

Ans: A

Explanation: Sq(3) = 9 and Sq(2) = 4, so 9 + 4 = 13.

Q110 | Syntax | VB 6.0 — Procedures

Where must `Optional` parameters appear in the parameter list?

A. Only as the first parameter
B. Before all required parameters
C. Anywhere in the list
D. After all required parameters

Ans: D

Explanation: Once a parameter is Optional, every parameter after it must also be Optional.

Q111 | Concept | VB 6.0 — Procedures

Which function checks whether an Optional Variant argument was passed?

A. IsEmpty
B. IsNull
C. IsMissing
D. IsObject

Ans: C

Explanation: `IsMissing(arg)` returns True if the caller omitted the Optional Variant argument.

Q112 | Tricky | VB 6.0 — Procedures

A function declared `Function Foo() As Integer` never assigns a value to `Foo`. What does it return?

A. -1
B. Null
C. A run-time error
D. 0

Ans: D

Explanation: The return value keeps its default (0 for numeric types, empty string for String).

Q113 | Error | VB 6.0 — Procedures

`Sub ShowMsg(s As String)` is called as `x = ShowMsg("Hi")`. What is the result?

A. Compile error: Expected function or variable
B. The Sub runs and x stays Empty
C. x receives "Hi"
D. x receives 0

Ans: A

Explanation: A Sub returns no value, so it cannot be used on the right-hand side of an assignment.

Q114 | Syntax | VB 6.0 — Procedures

Which statement leaves a Function immediately, before its `End Function`?

A. Exit Function
B. Exit Do
C. Return
D. Exit Sub

Ans: A

Explanation: Use `Exit Function` in a Function and `Exit Sub` in a Sub.

Q115 | Syntax | VB 6.0 — Procedures

Which keyword lets a procedure accept an arbitrary number of arguments?

A. Optional
B. Static
C. ParamArray
D. ByVal

Ans: C

Explanation: `ParamArray` must be the last parameter and is declared as a Variant array.

Q116 | Output | VB 6.0 — Procedures

What does the MsgBox display?

```
Function Fact(n As Integer) As Long
    If n <= 1 Then
        Fact = 1
    Else
        Fact = n * Fact(n - 1)
    End If
End Function

MsgBox Fact(4)
```

A. 10
B. 16
C. 24
D. 4

Ans: C

Explanation: Fact(4) = 4 × 3 × 2 × 1 = 24. The function calls itself (recursion) until n <= 1.

Q117 | Scenario | VB 6.0 — Procedures

A reusable routine must compute tax so it can be used inside an expression like `total = price + Tax(price)`. Which should be written?

A. A Function
B. A Type block
C. A Sub
D. A Static variable

Ans: A

Explanation: Only a Function returns a value that can be used within an expression.

Q118 | Concept | VB 6.0 — Procedures

Which feature lets you pass arguments in any order using the form `name:=value`?

A. ParamArray
B. Named arguments
C. Static binding
D. ByVal passing

Ans: B

Explanation: Example: `Greet greeting:="Hi", name:="Ravi"`.

Q119 | Output | VB 6.0 — Procedures

What does `Pw(5)` return?

```
Function Pw(b As Double, Optional e As Integer = 2) As Double
    Pw = b ^ e
End Function
```

A. 5
B. 0
C. 25
D. 10

Ans: C

Explanation: The Optional parameter `e` defaults to 2, so 5 ^ 2 = 25.

Q120 | Concept | VB 6.0 — Procedures

A procedure declared `Private` in a standard module can be called from:

A. Only procedures in the same module
B. Any procedure in the project
C. Only event procedures
D. Any form in the project

Ans: A

Explanation: Private limits a procedure’s scope to its own module. Public (the default) is project-wide.

Q121 | Output | VB 6.0 — Procedures

What does `Fib(6)` return?

```
Function Fib(n As Integer) As Long
    If n <= 2 Then
        Fib = 1
    Else
        Fib = Fib(n - 1) + Fib(n - 2)
    End If
End Function
```

A. 8
B. 5
C. 6
D. 13

Ans: A

Explanation: Fib(1) = Fib(2) = 1, then each value is the sum of the previous two: 1, 1, 2, 3, 5, 8.

Q122 | Error | VB 6.0 — Procedures

What happens when `Inc x` is compiled or run?

```
Sub Inc(n As Integer)
    n = n + 1
End Sub

Dim x As Long
Inc x
```

A. x is incremented normally
B. x is converted to Integer automatically
C. Compile error: ByRef argument type mismatch
D. Run-time error 6: Overflow

Ans: C

Explanation: A ByRef parameter needs an argument of exactly the same type. A Long variable cannot be passed to an Integer parameter.

Q123 | Concept | VB 6.0 — Procedures

If `As datatype` is omitted from a Function header, the function returns a:

A. Variant
B. String
C. Integer
D. Boolean

Ans: A

Explanation: Untyped declarations default to Variant in VB 6.0.

SECTION 8

Date Functions

Q124 - Q143
20 MCQs

Q124 | Concept | VB 6.0 — Date Functions

Which function returns the current system date AND time together?

A. Now
B. Time
C. Date
D. Timer

Ans: A

Explanation: `Date` returns only the date, `Time` only the time, and `Timer` the seconds elapsed since midnight.

Q125 | Concept | VB 6.0 — Date Functions

The `Timer` function returns:

A. The current date serial number
B. Seconds elapsed since midnight
C. The current time as text
D. Milliseconds since 1 January 1970

Ans: B

Explanation: Timer returns a Single and is commonly used to measure elapsed time.

Q126 | Output | VB 6.0 — Date Functions

What does `DateAdd("d", 10, #1/25/2024#)` return?

A. March 5, 2024
B. February 3, 2024
C. February 4, 2024
D. February 5, 2024

Ans: C

Explanation: 25 January + 10 days = 4 February (January has 31 days).

Q127 | Tricky | VB 6.0 — Date Functions

What does `DateAdd("m", 1, #1/31/2024#)` return?

A. March 1, 2024
B. February 29, 2024
C. February 28, 2024
D. March 2, 2024

Ans: B

Explanation: When the target month is shorter, VB returns the last valid day. 2024 is a leap year.

Q128 | Output | VB 6.0 — Date Functions

What does `DateDiff("d", #1/1/2024#, #3/1/2024#)` return?

A. 59
B. 60
C. 2
D. 61

Ans: B

Explanation: January has 31 days and February 2024 has 29 days (leap year): 31 + 29 = 60.

Q129 | Tricky | VB 6.0 — Date Functions

What does `DateDiff("yyyy", #12/31/2023#, #1/1/2024#)` return?

A. 1
B. 2
C. -1
D. 0

Ans: A

Explanation: DateDiff counts calendar boundaries crossed, not full years. One year boundary lies between the two dates.

Q130 | Output | VB 6.0 — Date Functions

What does `DatePart("q", #8/15/2024#)` return?

A. 8
B. 2
C. 3
D. 4

Ans: C

Explanation: The interval "q" gives the calendar quarter. August is in Q3.

Q131 | Tricky | VB 6.0 — Date Functions

What does `Weekday(#1/1/2024#)` return? (1 January 2024 was a Monday.)

A. 1
B. 2
C. 0
D. 7

Ans: B

Explanation: By default Sunday = 1, so Monday = 2.

Q132 | Tricky | VB 6.0 — Date Functions

What does `DateSerial(2024, 2, 30)` return?

A. Run-time error
B. March 1, 2024
C. March 2, 2024
D. February 29, 2024

Ans: B

Explanation: DateSerial rolls over out-of-range values. February 2024 has 29 days, so day 30 becomes 1 March.

Q133 | Syntax | VB 6.0 — Date Functions

Which interval string represents MINUTES in `DateAdd` and `DateDiff`?

A. "min"
B. "m"
C. "mi"
D. "n"

Ans: D

Explanation: "m" means month. Minutes use "n", hours "h" and seconds "s".

Q134 | Output | VB 6.0 — Date Functions

What does `Format(#3/5/2024#, "dd-mmm-yyyy")` return?

A. "05-03-2024"
B. "05-Mar-2024"
C. "5-March-2024"
D. "03-May-2024"

Ans: B

Explanation: `dd` = two-digit day, `mmm` = abbreviated month name, `yyyy` = four-digit year. Dates in # # are written month first.

Q135 | Concept | VB 6.0 — Date Functions

Which function tests whether an expression can be converted to a date?

A. IsNumeric
B. IsDate
C. DateValue
D. CDate

Ans: B

Explanation: `IsDate` returns True or False. `CDate` converts and raises an error if the value is invalid.

Q136 | Concept | VB 6.0 — Date Functions

Internally, the fractional part of a Date value represents:

A. The time of day
B. The year
C. The day of the week
D. The month

Ans: A

Explanation: The integer part is the date and the fractional part is the time. For example, 0.5 is 12:00 noon.

Q137 | Output | VB 6.0 — Date Functions

What does `Hour(#3:45:30 PM#)` return?

A. 45
B. 3
C. 15
D. 30

Ans: C

Explanation: Hour returns 0–23, so 3 PM is 15.

Q138 | Output | VB 6.0 — Date Functions

What does `MonthName(3)` return?

A. "3"
B. "May"
C. "March"
D. "Mar"

Ans: C

Explanation: `MonthName(n)` returns the full month name unless its Abbreviate argument is True.

Q139 | Tricky | VB 6.0 — Date Functions

What does `Hour(TimeSerial(10, 75, 0))` return?

A. 75 **B.** 10
C. 12 **D.** 11

Ans: D

Explanation: TimeSerial rolls the excess minutes over: 10:75 becomes 11:15, so the hour is 11.

Q140 | Output | VB 6.0 — Date Functions

What does `WeekdayName(Weekday(#1/1/2024#))` return?

A. "Tuesday" **B.** "Monday"
C. "Sunday" **D.** "Saturday"

Ans: B

Explanation: 1 January 2024 was a Monday. Weekday gives 2 and WeekdayName(2) is "Monday" (Sunday = 1).

Q141 | Tricky | VB 6.0 — Date Functions

What does `DateAdd("yyyy", -1, #2/29/2024#)` return?

A. February 28, 2023 **B.** February 27, 2023
C. February 29, 2023 **D.** March 1, 2023

Ans: A

Explanation: A negative number goes back in time. 2023 is not a leap year, so VB returns the last valid day of February.

Q142 | Output | VB 6.0 — Date Functions

What does `DatePart("y", #3/1/2024#)` return?

A. 62 **B.** 61
C. 3 **D.** 60

Ans: B

Explanation: "y" is the day of the year: 31 (Jan) + 29 (Feb, leap year) + 1 = 61.

Q143 | Output | VB 6.0 — Date Functions

What does `DateDiff("h", #1/1/2024 6:00:00 AM#, #1/2/2024 6:00:00 PM#)` return?

A. 24 **B.** 18
C. 12 **D.** 36

Ans: D

Explanation: From 6 AM on 1 January to 6 PM on 2 January is 24 + 12 = 36 hours.

SECTION 9

Mathematical Functions

Q144 - Q160
17 MCQs

Q144 | Tricky | VB 6.0 — Math Functions

What is `Int(-3.6) + Fix(-3.6)`?

A. -4
B. -6
C. -8
D. -7

Ans: D

Explanation: Int rounds down to -4. Fix truncates toward zero to -3. The sum is -7.

Q145 | Tricky | VB 6.0 — Math Functions

What is `Round(2.5) + Round(3.5)` in VB 6.0?

A. 5
B. 8
C. 6
D. 7

Ans: C

Explanation: Round uses banker's rounding (to the even number): Round(2.5) = 2 and Round(3.5) = 4.

Q146 | Concept | VB 6.0 — Math Functions

`Sgn(-25)` returns:

A. 0
B. 1
C. 25
D. -1

Ans: D

Explanation: Sgn returns -1 for negative, 0 for zero and 1 for positive numbers.

Q147 | Error | VB 6.0 — Math Functions

What happens when `Sqr(-4)` is evaluated?

A. Run-time error 5: Invalid procedure call or argument
B. It returns 2
C. It returns -2
D. It returns 0

Ans: A

Explanation: The square root of a negative number is not defined in VB 6.0, which raises an error.

Q148 | Tricky | VB 6.0 — Math Functions

What is the value of `-17 Mod 5`?

A. 2
B. -3
C. -2
D. 3

Ans: C

Explanation: The result of Mod takes the sign of the dividend, so -17 Mod 5 = -2.

Q149 | Output | VB 6.0 — Math Functions

What is the value of `7 / 2 + 7 \ 2 + 7 Mod 2`?

A. 7
B. 8
C. 6.5
D. 7.5

Ans: D

Explanation: 7 / 2 = 3.5 (floating-point), 7 \ 2 = 3 (integer division), 7 Mod 2 = 1. Total 7.5.

Q150 | Tricky | VB 6.0 — Math Functions

What is the value of `-2 ^ 2` in VB 6.0?

A. 0.25
B. -0.25
C. -4
D. 4

Ans: C

Explanation: The exponent operator has higher precedence than unary minus, so it is evaluated as -(2 ^ 2).

Q151 | Tricky | VB 6.0 — Math Functions

What is the value of `2 ^ 3 ^ 2`?

A. 64
B. 18
C. 128
D. 512

Ans: A

Explanation: Operators of equal precedence are evaluated left to right: (2 ^ 3) ^ 2 = 8 ^ 2 = 64.

Q152 | Syntax | VB 6.0 — Math Functions

Which expression simulates a throw of a die (whole numbers 1 to 6)?

A. Int(Rnd * 6)
B. Int(Rnd * 6) + 1
C. Rnd * 6
D. Int(Rnd * 7)

Ans: B

Explanation: Rnd returns a value from 0 up to but not including 1. Int(Rnd*6) gives 0–5, so adding 1 gives 1–6.

Q153 | Concept | VB 6.0 — Math Functions

What is the purpose of the `Randomize` statement?

A. It resets Rnd to 0
B. It returns a random integer
C. It sorts an array randomly
D. It seeds the random number generator so each run gives a different sequence

Ans: D

Explanation: Without Randomize, Rnd gives the same sequence every time the program runs.

Q154 | Concept | VB 6.0 — Math Functions

The `Log(x)` function in VB 6.0 returns:

A. The natural logarithm (base e) of x
B. The base-10 logarithm of x
C. The antilogarithm of x
D. The base-2 logarithm of x

Ans: A

Explanation: `Log` is the natural logarithm and `Exp` is its inverse. Base-10 needs Log(x) / Log(10).

Q155 | Concept | VB 6.0 — Math Functions

Angles used by `Sin`, `Cos`, `Tan` and `Atn` are measured in:

A. Degrees
B. Minutes
C. Radians
D. Grads

Ans: C

Explanation: To convert degrees to radians, multiply by π/180.

Q156 | Syntax | VB 6.0 — Math Functions

Which expression gives the value of π?

A. 4 * Sin(1)
B. 4 * Tan(1)
C. 4 * Atn(1)
D. 2 * Cos(1)

Ans: C

Explanation: Atn(1) = π/4, hence π = 4 × Atn(1).

Q157 | Concept | VB 6.0 — Math Functions

What does `Rnd(0)` return?

A. A brand-new random number
B. The most recently generated random number
C. The current seed value
D. Always 0

Ans: B

Explanation: Rnd(0) repeats the last number. Rnd with a positive or omitted argument gives the next number in the sequence.

Q158 | Output | VB 6.0 — Math Functions

What is `Exp(0) + Log(1)`?

A. Run-time error
B. 1
C. 2
D. 0

Ans: B

Explanation: Exp(0) = 1 and Log(1) = 0 (natural log), so the sum is 1.

Q159 | Tricky | VB 6.0 — Math Functions

What is the value of `7.6 Mod 2.4`?

A. 0
B. 0.4
C. 2.8
D. 1

Ans: A

Explanation: Mod rounds both operands to integers first: 8 Mod 2 = 0.

Q160 | Output | VB 6.0 — Math Functions

What is `Abs(-3) * Sgn(-9) + Fix(4.9)`?

A. 7
B. 4
C. -7
D. 1

Ans: D

Explanation: 3 × (-1) + 4 = 1.

SECTION 10

String Functions

Q161 – Q180
20 MCQs

Q161 | Output | VB 6.0 — String Functions

What does `Mid("COMPUTER", 4, 3)` return?

A. "UTE"
B. "PUT"
C. "MPU"
D. "OMP"

Ans: B

Explanation: Mid(string, start, length) takes 3 characters from position 4: P-U-T.

Q162 | Output | VB 6.0 — String Functions

What does `Left("EXAMINATION", 4) & Right("EXAMINATION", 3)` return?

A. "EXAMTION"
B. "XAMION"
C. "EXAMION"
D. "EXAIION"

Ans: C

Explanation: Left gives "EXAM" and Right gives "ION".

Q163 | Output | VB 6.0 — String Functions

What does `InStr("Hello World", "o")` return?

A. 5
B. 8
C. 0
D. 4

Ans: A

Explanation: InStr is 1-based and finds the first match, at position 5. (`InStrRev` would give 8.)

Q164 | Output | VB 6.0 — String Functions

What does `InStrRev("Hello World", "o")` return?

A. 8 **B.** 0
C. 2 **D.** 5

Ans: A

Explanation: InStrRev searches from the end of the string and returns the position of the last "o".

Q165 | Output | VB 6.0 — String Functions

What is `Len(Trim(" VB 6 "))`?

A. 6 **B.** 8
C. 4 **D.** 5

Ans: C

Explanation: Trim removes leading and trailing spaces, leaving "VB 6", which has 4 characters (the inner space counts).

Q166 | Output | VB 6.0 — String Functions

What does `StrReverse("VB6")` return?

A. "V6B" **B.** "6VB"
C. "6BV" **D.** "BV6"

Ans: C

Explanation: StrReverse reverses the order of the characters.

Q167 | Output | VB 6.0 — String Functions

What does `Replace("banana", "a", "o", 1, 2)` return?

A. "bonona" **B.** "bonana"
C. "banono" **D.** "bonono"

Ans: A

Explanation: The last argument limits replacement to the first 2 occurrences of "a".

Q168 | Output | VB 6.0 — String Functions

What does `Chr(Asc("A") + 2)` return?

A. "67" **B.** "A"
C. "C" **D.** "B"

Ans: C

Explanation: Asc("A") = 65, and Chr(67) is "C".

Q169 | Tricky | VB 6.0 — String Functions

What does `StrComp("apple", "Apple")` return with the default (binary) comparison?

A. 0
B. 1
C. -1
D. True

Ans: B

Explanation: Binary comparison is case-sensitive. "a" (97) is greater than "A" (65), so the first string is greater: 1.

Q170 | Tricky | VB 6.0 — String Functions

What does `MsgBox "5" + 3 & "2"` display?

A. 532
B. 82
C. Type mismatch error
D. 10

Ans: B

Explanation: `+` has higher precedence than `&`. "5" + 3 = 8 (numeric addition), then 8 & "2" = "82".

Q171 | Output | VB 6.0 — String Functions

What does `UBound(Split("a,b,c,d", ","))` return?

A. 4
B. 2
C. 3
D. 7

Ans: C

Explanation: Split returns a zero-based array of 4 elements (indexes 0 to 3).

Q172 | Tricky | VB 6.0 — String Functions

What is `Len(Space(3) & String(2, "AB"))`?

A. 6
B. 4
C. 7
D. 5

Ans: D

Explanation: Space(3) is 3 blanks. `String(2, "AB")` repeats only the first character, giving "AA". 3 + 2 = 5.

Q173 | Output | VB 6.0 — String Functions

What does `Mid("ABCDEF", 3)` return?

A. "CD"
B. "CDEF"
C. "DEF"
D. "ABC"

Ans: B

Explanation: When the length argument is omitted, Mid returns everything from the start position to the end.

Q174 | Syntax | VB 6.0 — String Functions

Which pattern character in the `Like` operator matches exactly ONE digit?

A. #
B. *
C. ?
D. !

Ans: A

Explanation: `#` matches a digit, `?` any single character and `*` any number of characters.

Q175 | Output | VB 6.0 — String Functions

What does `InStr(5, "banana bandana", "an")` return?

A. 2
B. 4
C. 12
D. 9

Ans: D

Explanation: The search starts at position 5, skipping the matches at positions 2 and 4. The next "an" starts at position 9.

Q176 | Output | VB 6.0 — String Functions

What does `"[" & RTrim(" Hi ") & "]"` return?

A. "[Hi]"
B. "[Hi]"
C. "[Hi]"
D. "[Hi]"

Ans: B

Explanation: RTrim removes only trailing spaces. The leading spaces remain.

Q177 | Concept | VB 6.0 — String Functions

The `$` versions of string functions (for example `Left$` and `Mid$`) return:

A. Only uppercase text
B. An array of characters
C. A Null-capable Variant
D. A String data type, not a Variant

Ans: D

Explanation: `Left` returns a Variant holding a String, while `Left$` returns a true String.

Q178 | Output | VB 6.0 — String Functions

What is the value of `s` after this code?

```
s = "HELLO"
Mid(s, 1, 1) = "J"
```

A. "J"
B. "JHELLO"
C. "JELLO"
D. "HELLO"

Ans: C

Explanation: The `Mid` statement overwrites characters in place: 1 character from position 1 is replaced by "J".

Q179 | Output | VB 6.0 — String Functions

What does `Len("A" & vbCrLf & "B")` return?

A. 5
B. 4
C. 3
D. 2

Ans: B

Explanation: `vbCrLf` is two characters (Chr(13) + Chr(10)), so 1 + 2 + 1 = 4.

Q180 | Tricky | VB 6.0 — String Functions

With binary comparison, what does `"Z" < "a"` return?

A. Null
B. False
C. True
D. Run-time error

Ans: C

Explanation: Comparison uses character codes. "Z" is 90 and "a" is 97, so all uppercase letters sort before lowercase ones.

SECTION 11

Conversion Functions

Q181 - Q198
18 MCQs

Q181 | Tricky | VB 6.0 — Conversion

After `Dim n As Integer` and `n = 3.7`, the value stored in `n` is:

A. 4
B. 3
C. 3.7
D. Run-time error

Ans: A

Explanation: Assigning a fractional value to an Integer rounds it (it does not truncate).

Q182 | Error | VB 6.0 — Conversion

What happens when `CInt(40000)` is evaluated?

A. Run-time error 13: Type mismatch
B. Run-time error 6: Overflow
C. It returns 40000
D. It returns 32767

Ans: B

Explanation: Integer holds at most 32,767, so 40000 overflows. `CLng(40000)` works.

Q183 | Tricky | VB 6.0 — Conversion

What is `Val("12abc") + Val("abc12")`?

A. 0
B. 24
C. 12
D. Type mismatch error

Ans: C

Explanation: Val reads digits until it meets a non-numeric character. "12abc" gives 12 and "abc12" gives 0.

Q184 | Output | VB 6.0 — Conversion

What does `Val("1,234")` return?

A. 1234
B. 1
C. 1.234
D. Run-time error

Ans: B

Explanation: Val does not recognise the comma as part of a number, so it stops reading at the comma.

Q185 | Tricky | VB 6.0 — Conversion

What is `Len(Str(25)) - Len(CStr(25))`?

A. 0 **B.** 1
C. -1 **D.** 2

Ans: B

Explanation: Str(25) = " 25" (leading space reserved for the sign), but CStr(25) = "25". 3 − 2 = 1.

Q186 | Concept | VB 6.0 — Conversion

What does `CBool(-3)` return?

A. Run-time error **B.** -3
C. False **D.** True

Ans: D

Explanation: CBool converts 0 to False and any non-zero number to True.

Q187 | Tricky | VB 6.0 — Conversion

What does `TypeName(7 / 2)` return?

A. "Integer" **B.** "Single"
C. "Variant" **D.** "Double"

Ans: D

Explanation: The `/` operator always yields a floating-point result (Double). `7 \ 2` would be an Integer.

Q188 | Tricky | VB 6.0 — Conversion

What does `TypeName(40000)` return?

A. "Integer" **B.** "Long"
C. "Double" **D.** "Single"

Ans: B

Explanation: An integer literal beyond 32,767 is automatically treated as Long.

Q189 | Output | VB 6.0 — Conversion

What does `Hex(255)` return?

A. "FE" **B.** "&HFF"
C. "FF" **D.** "F0"

Ans: C

Explanation: Hex returns a String of hexadecimal digits without a prefix. 255 = FF.

Q190 | Output | VB 6.0 — Conversion

What does `StrConv("visual basic", vbProperCase)` return?

A. "Visual basic" **B.** "VISUAL BASIC"
C. "visual basic" **D.** "Visual Basic"

Ans: D

Explanation: vbProperCase capitalises the first letter of each word.

Q191 | Concept | VB 6.0 — Conversion

Which of these calls returns False?

A. IsNumeric("12.5")
B. IsNumeric("1E3")
C. IsNumeric("-7")
D. IsNumeric("12abc")

Ans: D

Explanation: IsNumeric returns True only if the entire expression can be read as a number. Scientific notation such as 1E3 qualifies.

Q192 | Error | VB 6.0 — Conversion

What is the result of `CInt("abc")`?

A. Run-time error 6: Overflow
B. Run-time error 13: Type mismatch
C. 0
D. An empty string

Ans: B

Explanation: The C-conversion functions raise an error for non-numeric text, unlike Val, which returns 0.

Q193 | Output | VB 6.0 — Conversion

What does `CCur(2.123456)` return?

A. 2.1235
B. 2.123456
C. 2.1234
D. 2.12

Ans: A

Explanation: Currency holds four decimal places, so the value is rounded to 2.1235.

Q194 | Output | VB 6.0 — Conversion

What does `Month(CDate("March 15, 2024"))` return?

A. 1
B. 15
C. 3
D. 2024

Ans: C

Explanation: CDate converts the text to a Date, and Month returns the month number.

Q195 | Concept | VB 6.0 — Conversion

Which function removes the fractional part of a number by truncating toward zero?

A. CInt
B. Fix
C. CLng
D. Int

Ans: B

Explanation: Fix(-4.7) = -4. Int rounds down (-5), while CInt and CLng round to the nearest integer.

Q196 | Concept | VB 6.0 — Conversion

Which function tests whether a variable holds an array?

A. IsArray
B. IsObject
C. IsMissing
D. IsNull

Ans: A

Explanation: `IsArray(v)` returns True if v is an array.

Q197 | Tricky | VB 6.0 — Conversion

What does `IsNull("")` return?

A. True
B. Null
C. Run-time error
D. False

Ans: D

Explanation: An empty string is a valid string value of length zero. Null means that no valid data exists.

Q198 | Output | VB 6.0 — Conversion

What does `CInt("3.7") + Val("3.7")` return?

A. 6.7
B. 7.7
C. 7
D. 8

Ans: B

Explanation: CInt rounds 3.7 to 4, while Val keeps 3.7. 4 + 3.7 = 7.7.

SECTION 12

Important and Tricky Questions

Q199 – Q216
18 MCQs

Q199 | Match | VB 6.0 — Data Types

Match the type-declaration characters (List I) with the data types (List II).
List I: 1. % | 2. & | 3. # | 4. $
List II: a. Integer | b. Long | c. Double | d. String

A. 1-a, 2-b, 3-c, 4-d
B. 1-b, 2-a, 3-c, 4-d
C. 1-a, 2-b, 3-d, 4-c
D. 1-c, 2-d, 3-a, 4-b

Ans: A

Explanation: % is Integer, & is Long, # is Double and $ is String.

Q200 | Match | VB 6.0 — Control Flow

Match the statement/function (List I) with its feature (List II).
List I: 1. IIf | 2. Select Case | 3. For Each | 4. Exit Do
List II: a. Multi-way branch on one expression | b. Visits each element of a collection | c. Inline function evaluating both result parts | d. Leaves a Do loop early

A. 1-b, 2-a, 3-c, 4-d
B. 1-c, 2-a, 3-b, 4-d
C. 1-c, 2-b, 3-a, 4-d
D. 1-a, 2-c, 3-b, 4-d

Ans: B

Explanation: IIf is an inline function (c), Select Case branches on one expression (a), For Each traverses elements (b) and Exit Do leaves a Do loop (d).

Q201 | Match | VB 6.0 — Math Functions

Match each expression (List I) with its value (List II).
List I: 1. Sgn(-2.5) | 2. Fix(-2.5) | 3. Abs(-2.5) | 4. Int(-2.5)
List II: a. -3 | b. 2.5 | c. -2 | d. -1

A. 1-d, 2-a, 3-b, 4-c
B. 1-d, 2-c, 3-b, 4-a
C. 1-c, 2-d, 3-b, 4-a
D. 1-d, 2-c, 3-a, 4-b

Ans: B

Explanation: Sgn gives -1; Fix truncates to -2; Abs gives 2.5; Int rounds down to -3.

Q202 | Match | VB 6.0 — Scope

Match the declaration (List I) with its scope/lifetime (List II).
List I: 1. Dim inside a procedure | 2. Dim in a form's Declarations section | 3. Public in a standard module | 4. Static inside a procedure
List II: a. Entire project | b. Local; lost when the procedure ends | c. Module-level | d. Local; value retained between calls

A. 1-b, 2-c, 3-a, 4-d
B. 1-b, 2-a, 3-c, 4-d
C. 1-d, 2-c, 3-a, 4-b
D. 1-c, 2-b, 3-a, 4-d

Ans: A

Explanation: Dim in a procedure is local and temporary (b). Dim in a Declarations section is module-level (c). Public in a standard module is project-wide (a). Static keeps its value (d).

Q203 | Match | VB 6.0 — Date Functions

Match the date function (List I) with its purpose (List II).
List I: 1. Now | 2. DateAdd | 3. DateDiff | 4. DatePart
List II: a. Interval between two dates | b. Current date and time | c. Adds an interval to a date | d. Returns one part of a date

A. 1-b, 2-c, 3-d, 4-a
B. 1-c, 2-b, 3-a, 4-d
C. 1-b, 2-a, 3-c, 4-d
D. 1-b, 2-c, 3-a, 4-d

Ans: D

Explanation: Now (b), DateAdd (c), DateDiff (a) and DatePart (d).

Q204 | Assertion–Reason | VB 6.0 — Scope

Assertion (A): A `Static` variable keeps its value between calls to its procedure.
Reason (R): A `Static` variable is visible to all procedures in the module.

A. Both A and R are true, and R is the correct explanation of A
B. Both A and R are true, but R is not the correct explanation of A
C. A is true, but R is false
D. A is false, but R is true

Ans: C

Explanation: A is true. R is false because a Static variable keeps local scope; only its lifetime is extended.

Q205 | Statement | VB 6.0 — Arrays

Statement I: `Dim a(5)` creates six elements by default.
Statement II: `UBound(a)` returns the total number of elements in the array.

A. Only Statement I is true
B. Only Statement II is true
C. Both Statements I and II are true
D. Neither Statement I nor II is true

Ans: A

Explanation: I is true (indexes 0–5). II is false: UBound returns the highest index (5), not the count.

Q206 | Assertion–Reason | VB 6.0 — Procedures

Assertion (A): A Sub procedure returns a value through its own name.
Reason (R): Only Function procedures can be used inside expressions to supply a value.

A. Both A and R are true, and R is the correct explanation of A
B. Both A and R are true, but R is not the correct explanation of A
C. A is true, but R is false
D. A is false, but R is true

Ans: D

Explanation: A is false: only a Function returns a value via its name. R is true.

Q207 | Statement | VB 6.0 — Procedures

Statement I: `Exit Function` can be used inside a Function procedure.
Statement II: A Function returns its result by assigning a value to the function name.

A. Only Statement I is true
B. Only Statement II is true
C. Both Statements I and II are true
D. Neither Statement I nor II is true

Ans: C

Explanation: Both are correct. Exit Function leaves early, with the last value assigned to the name returned.

Q208 | Assertion–Reason | VB 6.0 — Loops

Assertion (A): `For Each` can be used to traverse the elements of an array.
Reason (R): The control variable of a `For Each` loop over an array must be of the Variant type.

A. Both A and R are true, and R is the correct explanation of A
B. Both A and R are true, but R is not the correct explanation of A
C. A is true, but R is false
D. A is false, but R is true

Ans: B

Explanation: Both statements are true, but R is a rule about the control variable. It does not explain why For Each works on arrays.

Q209 | Statement | VB 6.0 — Control Flow

Statement I: `IIf` is a block statement closed with `End IIf`.
Statement II: `Select Case` is closed with `End If`.

A. Only Statement I is true
B. Only Statement II is true
C. Both Statements I and II are true
D. Neither Statement I nor II is true

Ans: D

Explanation: IIf is a single-line function and Select Case ends with `End Select`.

Q210 | Assertion–Reason | VB 6.0 — Control Flow

Assertion (A): `IIf` can raise a division-by-zero error even when the condition selects the safe branch.
Reason (R): `IIf` evaluates both its truepart and falsepart before returning one.

A. Both A and R are true, and R is the correct explanation of A
B. Both A and R are true, but R is not the correct explanation of A
C. A is true, but R is false
D. A is false, but R is true

Ans: A

Explanation: Both parts are always evaluated, so `IIf(n <> 0, 100 / n, 0)` fails when n = 0.

Q211 | Statement | VB 6.0 — Control Flow

Statement I: In `Select Case`, execution falls through into the next Case block after a match.
Statement II: The `Case Else` clause is optional.

A. Only Statement I is true
B. Only Statement II is true
C. Both Statements I and II are true
D. Neither Statement I nor II is true

Ans: B

Explanation: VB 6.0 never falls through. Only the first matching Case runs. Case Else is optional.

Q212 | Error | VB 6.0 — Constants

What happens when the following code is compiled or run?

```
Const MAX = 10
MAX = 20
```

A. Compile error: Assignment to constant not permitted
B. Run-time error 6: Overflow
C. MAX becomes 20
D. MAX is ignored silently

Ans: A

Explanation: A constant declared with `Const` cannot be assigned a new value.

Q213 | Error | VB 6.0 — Scope

What is wrong with this code?

```
Sub Test()
    Public x As Integer
End Sub
```

A. Test must be declared Static
B. Integer needs a type suffix
C. A Sub cannot contain declarations
D. Public cannot be used to declare a variable inside a procedure

Ans: D

Explanation: Inside a procedure only `Dim` and `Static` are allowed. `Public` and `Private` belong to the Declarations section.

Q214 | Scenario | VB 6.0 — Strings and Loops

To examine every character of the string `s` one at a time, which loop is correct?

A. `For i = 1 To Len(s)` using `Right(s, i)`
B. `For i = 0 To Len(s)` using `Mid(s, i, 1)`
C. `For Each c In s` using `Left(c, 1)`
D. `For i = 1 To Len(s)` using `Mid(s, i, 1)`

Ans: D

Explanation: String positions in VB 6.0 start at 1. `Mid(s, 0, 1)` raises run-time error 5, and For Each cannot iterate over characters of a String.

Q215 | Scenario | VB 6.0 — Control Flow

Converting marks into grades (A, B, C, D, F) with several ranges is most readable using:

A. IIf with a Boolean flag
B. Select Case with `Case Is >= …` clauses
C. For Each over the marks
D. A Static variable

Ans: B

Explanation: Select Case handles several ranges in one clear block, and a nested IIf quickly becomes unreadable.

Q216 | Tricky | VB 6.0 — Operators

What is the value of the following expression?

```
2 + 3 * 4 ^ 2 \ 5 Mod 3
```

A. 9
B. 11
C. 2
D. 5

Ans: C

Explanation: Precedence: ^, then *, then \, then Mod, then +. 4^2 = 16, 3*16 = 48, 48 \ 5 = 9, 9 Mod 3 = 0, and 2 + 0 = 2.

SECTION 13

Mixed OAVS TGT Practice Set

Q217 – Q230
14 MCQs

Q217 | Output | VB 6.0 — Loops and Operators

What does the MsgBox display?

```
For i = 1 To 10
    If i Mod 2 = 0 Then s = s + i
Next i
MsgBox s
```

A. 55
B. 30
C. 20
D. 25

Ans: B

Explanation: Only the even numbers 2 + 4 + 6 + 8 + 10 are added, giving 30.

Q218 | Output | VB 6.0 — Functions and Strings

What does the MsgBox display?

```
Function Rev(s As String) As String
    Dim i As Integer
    For i = Len(s) To 1 Step -1
        Rev = Rev & Mid(s, i, 1)
    Next i
End Function

MsgBox Rev("OAVS")
```

A. "SVAO"
B. "SAVO"
C. "OAVS"
D. "VAOS"

Ans: A

Explanation: The loop walks from the last character to the first, building the reversed string.

Q219 | Output | VB 6.0 — Arrays and Loops

What does the MsgBox display?

```
Dim a As Variant, m As Integer, i As Integer
a = Array(4, 9, 2, 7)
m = a(0)
For i = 1 To UBound(a)
    If a(i) > m Then m = a(i)
Next i
MsgBox m
```

A. 2
B. 9
C. 4
D. 7

Ans: B

Explanation: The loop finds the largest element of the array: 9.

Q220 | Output | VB 6.0 — Nested Loops

What is the final value of `s`?

```
For i = 1 To 2
    For j = 1 To 3
        s = s & i * j & " "
    Next j
Next i
```

A. "1 2 3 4 5 6 "
B. "1 2 3 2 4 6 "
C. "1 4 9 "
D. "1 2 2 4 3 6 "

Ans: B

Explanation: The product i × j is appended. i = 1 gives 1 2 3 and i = 2 gives 2 4 6.

Q221 | Output | VB 6.0 — Select Case

What is the value of `r`?

```
c = "b"
Select Case UCase(c)
    Case "A", "E": r = 1
    Case "B" To "D": r = 2
    Case Else: r = 3
End Select
```

A. 1
B. 0
C. 3
D. 2

Ans: D

Explanation: UCase("b") = "B", which is not in "A","E" but lies in the range "B" To "D".

Q222 | Output | VB 6.0 — IIf

What does the expression below return?

```
IIf(5 > 3, IIf(2 > 4, "A", "B"), "C")
```

A. "C"
B. "A"
C. True
D. "B"

Ans: D

Explanation: 5 > 3 is True, so the inner IIf is evaluated. 2 > 4 is False, so it gives "B".

Q223 | Scenario | VB 6.0 — Math Functions

Which expression rounds 3.14159 to two decimal places?

A. Round(3.14159, 2)
B. CInt(3.14159, 2)
C. Fix(3.14159 * 100)
D. Int(3.14159, 2)

Ans: A

Explanation: The second argument of Round gives the number of decimal places. Int and CInt accept only one argument.

Q224 | Output | VB 6.0 — String Functions

What does the MsgBox display?

```
f = "report.final.txt"
MsgBox Right(f, Len(f) - InStrRev(f, "."))
```

A. "final.txt"
B. "report"
C. "final"
D. "txt"

Ans: D

Explanation: InStrRev finds the last "." (position 13). Right takes the remaining 16 − 13 = 3 characters.

Q225 | Error | VB 6.0 — Functions

What is wrong with the last statement?

```
Function Add(a As Integer, b As Integer) As Integer
    Add = a + b
End Function

Dim r As Integer
r = Add 3, 4
```

A. r must be declared as Variant
B. ByVal is missing from the parameters
C. Add must be declared as a Sub
D. Arguments must be enclosed in parentheses when the function value is used

Ans: D

Explanation: `Add 3, 4` is valid only for a call without using the return value. Assignment needs `r = Add(3, 4)`.

Q226 | Output | VB 6.0 — Procedures

What does the MsgBox display?

```
Sub Swap(a As Integer, b As Integer)
    Dim t As Integer
    t = a: a = b: b = t
End Sub

x = 3: y = 8
Swap x, y
MsgBox x - y
```

A. -5
B. 5
C. 11
D. 0

Ans: B

Explanation: Arguments are passed ByRef by default, so x and y are swapped (x = 8, y = 3). Then x − y = 5.

Q227 | Output | VB 6.0 — Strings and Conversion

What does the expression below return?

```
Val(Mid("AB123XY", 3, 3)) + 1
```

A. 1231
B. Run-time error
C. 124
D. 24

Ans: C

Explanation: Mid gives "123", Val converts it to 123, and adding 1 gives 124.

Q228 | Output | VB 6.0 — 2-D Arrays

What does the MsgBox display?

```
Dim m(1 To 2, 1 To 2) As Integer
Dim i As Integer, j As Integer
For i = 1 To 2
    For j = 1 To 2
        m(i, j) = i + j
    Next j
Next i
MsgBox m(2, 2) + m(1, 2)
```

A. 8
B. 6
C. 5
D. 7

Ans: D

Explanation: m(i, j) = i + j, so m(2,2) = 4 and m(1,2) = 3. Their sum is 7.

Q229 | Output | VB 6.0 — Variant

What does the MsgBox display?

```
Dim v As Variant
MsgBox IsEmpty(v)
```

A. True
B. False
C. Run-time error
D. Null

Ans: A

Explanation: A Variant that has never been assigned a value holds the special value Empty.

Q230 | Tricky | VB 6.0 — Date Functions

What does the following expression return?

```
Day(DateSerial(2024, 3, 0))
```

A. 29
B. 31
C. 28
D. 30

Ans: A

Explanation: Day 0 of March is the last day of February. 2024 is a leap year, so February has 29 days.

Quick Answer Key

1 **B**	2 **D**	3 **A**	4 **A**	5 **D**	6 **C**	7 **A**	8 **A**	9 **B**	10 **A**
11 **D**	12 **C**	13 **A**	14 **C**	15 **C**	16 **B**	17 **C**	18 **A**	19 **D**	20 **D**
21 **B**	22 **C**	23 **B**	24 **A**	25 **C**	26 **A**	27 **B**	28 **B**	29 **D**	30 **B**
31 **A**	32 **A**	33 **B**	34 **A**	35 **C**	36 **C**	37 **B**	38 **C**	39 **A**	40 **D**
41 **A**	42 **B**	43 **D**	44 **C**	45 **A**	46 **B**	47 **D**	48 **A**	49 **B**	50 **C**
51 **C**	52 **A**	53 **D**	54 **B**	55 **C**	56 **C**	57 **A**	58 **A**	59 **B**	60 **C**
61 **A**	62 **D**	63 **D**	64 **A**	65 **C**	66 **B**	67 **D**	68 **A**	69 **D**	70 **B**
71 **C**	72 **B**	73 **D**	74 **A**	75 **A**	76 **B**	77 **D**	78 **D**	79 **C**	80 **D**
81 **B**	82 **D**	83 **C**	84 **C**	85 **A**	86 **C**	87 **C**	88 **D**	89 **C**	90 **A**
91 **D**	92 **B**	93 **D**	94 **C**	95 **A**	96 **D**	97 **C**	98 **D**	99 **D**	100 **C**
101 **D**	102 **B**	103 **C**	104 **D**	105 **D**	106 **B**	107 **D**	108 **C**	109 **A**	110 **D**
111 **C**	112 **D**	113 **A**	114 **A**	115 **C**	116 **C**	117 **A**	118 **B**	119 **C**	120 **A**
121 **A**	122 **C**	123 **A**	124 **A**	125 **B**	126 **C**	127 **B**	128 **B**	129 **A**	130 **C**
131 **B**	132 **B**	133 **D**	134 **B**	135 **B**	136 **A**	137 **C**	138 **C**	139 **D**	140 **B**
141 **A**	142 **B**	143 **D**	144 **D**	145 **C**	146 **D**	147 **A**	148 **C**	149 **D**	150 **C**
151 **A**	152 **B**	153 **D**	154 **A**	155 **C**	156 **C**	157 **B**	158 **B**	159 **A**	160 **D**
161 **B**	162 **C**	163 **A**	164 **A**	165 **C**	166 **C**	167 **A**	168 **C**	169 **B**	170 **B**
171 **C**	172 **D**	173 **B**	174 **A**	175 **D**	176 **B**	177 **D**	178 **C**	179 **B**	180 **C**
181 **A**	182 **B**	183 **C**	184 **B**	185 **B**	186 **D**	187 **D**	188 **B**	189 **C**	190 **D**
191 **D**	192 **B**	193 **A**	194 **C**	195 **B**	196 **A**	197 **D**	198 **B**	199 **A**	200 **B**
201 **B**	202 **A**	203 **D**	204 **C**	205 **A**	206 **D**	207 **C**	208 **B**	209 **D**	210 **A**
211 **B**	212 **A**	213 **D**	214 **D**	215 **B**	216 **C**	217 **B**	218 **A**	219 **B**	220 **B**
221 **D**	222 **D**	223 **A**	224 **D**	225 **D**	226 **B**	227 **C**	228 **D**	229 **A**	230 **A**